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IS 310

Culture as Data Final Essay

Introduction

At the beginning of this Culture as Data project, the angle I was approaching Spotify datasets from was extremely simple and purely just from a data science perspective. I wanted to understand what makes songs popular by looking at features like danceability, energy, duration, and chart rankings. As a result, I was treating all of these metrics as pretty objective measurements that could help explain why certain songs succeed on streaming platforms. However, after receiving lots of feedback and continuing to work with the dataset throughout the semester, I realized that this approach is inherently flawed to begin with. The issue is that Spotify's metrics are not accurate representations of music. They are internal interpretations that have been calculated by Spotify.

After this, I turned the project more into an audit of how streaming platforms classify music numerically and what their ideas of popularity and emotion are. This was no longer a "what makes songs popular" project, it was how streaming platforms even categorize popularity beyond just streams. Instead of just analyzing metrics like danceability, valence, energy etc., I was wondering how these calculations came to be. Most importantly, I was curious as to what cultural complexities get lost when something subjective like music becomes objective numerical data.

For my final dataset, I created a dataset of 500 songs using the Spotify 2023 Kaggle dataset as my base. I added additional layers to it including popularity scores, mood classifications, visibility categories, and audit annotations. By doing so, I was looking through the differences between computational and human classification / interpretation. I was also analyzing how platforms shape musical visibility and cultural consumption. This included contextual visibility categories to account for external influences such as playlist amplification and social media virality that impact streaming popularity beyond just the audio alone.

Over time, I also learned that streaming platforms do not simply measure musical culture. Instead, they actively participate in constructing it through recommendation systems, playlist infrastructures, and more.

The Dataset

This project is mainly based on a “Most Streamed Spotify Songs” 2023 dataset from Kaggle. The dataset has information about popular songs on streaming platforms in 2023, such as stream counts, playlist appearances, chart rankings, and audio features generated by Spotify. The original dataset has variables like:

- Danceability
- Energy
- Valence
- Acousticness
- Speechiness
- Instrumentalness
- BPM
- Spotify playlist counts
- Spotify chart rankings
- Apple Music chart rankings
- Deezer chart rankings
- Shazam chart rankings

The dataset was made by combining publicly available streaming and chart information from multiple music platforms. However, one of the first issues present was that a lot of the features were not transparent in how they were collected. Spotify does give general explanations for metrics like danceability or valence, but the computational process behind them is not publicly accessible.

This became one of the main limitations throughout my project. Even though the dataset looks very quantitative, many of the measurements are quite subjective and up to interpretation rather than objective. For example, Spotify describes “danceability” as a measurement of how suitable a song is for dancing based on things like tempo and beat strength. But these are still assumptions about music that can vary between cultures and personal interpretation. Similarly, “valence” tries to estimate how emotionally positive a song is, even though emotional interpretation in music is very subjective and tends to need context. Ultimately, Spotify’s metrics are not just natural facts about songs. The numbers are created through Spotify’s own machine learning systems and engineering decisions.

Computational Augmentation and Expanding

My initial submission only contained a manually sampled dataset of 75 songs. While this helped me look closer at the dataset, it did not meaningfully augment or transform the dataset itself. For the final version of the project, I expanded the dataset to 500 songs and introduced several new manually then computationally generated columns to audit and compare with Spotify's current values.

The new columns included:

- popularity_score
- spotify_mood_label
- interpreted_song_type
- mood_complexity
- platform_visibility_type
- external_visibility_factor
- audit_note

The popularity_score column was made to look at popularity as a platform constructed metric rather than just a singular objective measure like streams. I combined streams, playlist appearances, chart placements, and visibility indicators across multiple platforms including Spotify, Apple Music, Deezer, and Shazam. This taught me that popularity is multidimensional. One song can be streamed a lot but not on the charts while another song could go viral on social media but be on barely any playlists. Different platforms measure cultural success differently.

I also created the spotify_mood_label column using Spotify metrics like valence and energy. I then contrasted it with my own interpretive category called interpreted_song_type. These labels were put in in hope to put some more emotional nuance back into the dataset.

For example:

- upbeat pop anthem
- emotionally tense
- reflective acoustic mood
- danceable but emotionally negative
- mixed emotional tone

This became one of the most important parts of my project because it showed how computational systems can reduce emotional complexity. For instance, a song could have high danceability and high energy while still having sad lyrics. Spotify could classify it as upbeat, whereas we as listeners know it's very different. I also introduced a mood_complexity column to point out songs where computational metrics were contradictory. This became another way to see where numerical systems failed to capture the ambiguity of musical experience.

The platform_visibility_type column categorized songs according to how their popularity appeared to be constructed whether it was:

- playlist-driven

- chart-driven
- viral/social-media-driven
- cross-platform-balanced

This helped identify the structures that make success possible.

Lastly, I created an external_visibility_factor column which was used to help provide context about factors influencing popularity outside of the music itself on Spotify. This column tried to consider some other factors influencing streams and popularity, such as playlist boosts, virality on social media platforms, artist collaborations, and chart trends. Though there are clearly other factors involved in each of these categories, it was an attempt to move the project beyond focusing on music itself.

How Computation Played a Role

Computation was important in all stages of the project. I used the pandas library in Python for cleaning the data set, processing numeric data, calculating new variables, and labeling them. Using computational methods, I could increase the data set from 75 to 500 songs without changing the category system.

It was a good practice to see that computation is never neutral. During the computational augmentation process, I had to make interpretive choices throughout each stage. This became evident particularly in trying to estimate externally constructed systems like virality and playlist amplification through computation, which are influenced by broader contexts beyond classification. These included questions of:

- what counts as "upbeat"
- what thresholds define emotional positivity
- how to operationalize popularity
- how to classify platform visibility
- how to identify contradictions between metrics

Despite the fact that these categories had been created through computer programming, there was still some subjectivity based on my own presuppositions. This was particularly evident when I started to create the category interpreted_song_type. Even though this category was created via computational logic, the logic itself was still subjective.

Thus, computation does not get rid of human subjectivity in the process. Datasets might seem objective, but there is always some underlying subjectivity when it comes to categorization.

How Scale Shaped the Data

Another major theme throughout this project was the relationship between scale and nuance. My original 75-song dataset allowed for close reading and individual inspection. I could manually think about individual songs' moods and think about whether or not the values aligned with my own interpretations.

However, once the dataset scaled to 500 songs, this kind of close interpretive work became more difficult. To handle the larger dataset, I relied more on automated categorization systems and rule-based logic. While this made large-scale analysis possible, it also meant simplifying the data once again.

For example, songs with very different emotional tones could still end up grouped together under labels like "mixed emotional tone" or "upbeat pop anthem." It showed me that scaling data often requires reducing complexity. This was even clearer in the case of external visibility, since phenomena such as virality or cultural momentum tend to be very fluid and hard to define in a computerized system for several hundred songs.

This problem became especially evident when considering emotion and mood. Music is very individual, contextual, and cultural. Any attempt to convert this experience into numbers necessarily involves a loss of meaning and ambiguity.

At the same time, scaling also highlighted patterns that would otherwise be hard to detect manually. For example, several popular songs had really similar visibility frameworks, like gaining visibility through playlists. This implied that a song's stream amount might not just be because of the music but because of what playlists it ended up on. Hence, scaling helped discover patterns while also limiting their interpretation.

Ethical Considerations and Limitations

There are several key limitations to consider when examining this project. For example, the data set is platform-specific, since this project utilizes metrics from Spotify. The focus on this one platform limits the scope of what we are observing in terms of music culture.

This creates several biases:

- Songs popular outside streaming ecosystems may be excluded
- Certain genres may receive more playlist support than others
- English-language music may be overrepresented
- Platform recommendation systems influence visibility
- Streaming metrics privilege measurable engagement over cultural significance

Furthermore, the computational measures used by Spotify are closed and secretive. Due to lack of transparency regarding how computational attributes like danceability and valence were determined, users do not have any opportunity to check whether or not their classification is accurate. One of the limitations that then came along with my augmentation was the fact that my classification of songs brought extra subjectivity to the process.

Additionally, I was not able to incorporate the feedback about directly measuring real-world marketing campaigns, TikTok trends, fan communities, or promotional budgets. Instead, the external_visibility_factor column acts as more of a proxy based on available platform metrics such as playlist counts and chart rankings. Many of the most influential social and cultural factors behind popularity are still difficult to quantify computationally.

I also considered the ethics of computationally classifying emotional experiences. Music often carries deeply personal meanings to people that just cannot be translated into numbers or labels.

Scholarship Situating

This project was heavily influenced by scholarly articles in platform and media studies, as well as music recommendation systems.

Seaver's Computing Taste is especially insightful in its approach to recommendation systems as cultural systems. According to Seaver, recommendation systems are not just mechanisms for predicting users' tastes, they create and define those tastes. Such a viewpoint had a major impact on the way in which I looked at Spotify, from being a neutral platform where listener preferences are reflected, to one in which Spotify participates in constructing taste in music culture.

Also, Morris's research into music recommendation systems and their relationship to curation and playlisting is especially relevant to my research, as his ideas directly informed the platform_visibility_type categories that I created.

Conclusion

My project began as an initial effort to explore popular Spotify songs by looking at audio characteristics and streaming statistics. But as the semester wore on, I found that the real focus was on the auditing of how streaming platforms compute their notions of popularity, moods, and visibility. Instead of focusing solely on the characteristics of the listeners themselves, through data augmentation, I was able to understand the impact of platform algorithms and infrastructure on how users experience the playlist of songs and recommendations.

The one thing that really struck me about this project is that data is never found; it always involves interpretation and assumptions. Although it was computational methods that enabled me to scale up my dataset from 75 to 500 songs, I also lost out on emotional nuances and cultural considerations along the way. In all, I came to realize that music as data goes beyond analyzing songs but also entails understanding the platform that makes those songs popular.

References

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